

Extra — Creating Dictionaries: Facts With Names

Topic: Making a Dictionary · **Course:** Extra Worksheet · **Time:** about 30 minutes

A list keeps things in a row, and you reach them by number. A **dictionary** keeps facts with **names**, so you reach them by name instead of counting. On paper you will read dictionary code, predict it, fix it, and explain it in your own words.

Keep these words handy: **dictionary**, **key**, **value**, **{ }**, **key:value pair**.

1 · Predict 🧠

Read each snippet. Write what you think it will print.

```
player = {"name": "Mia", "level": 3}
print(player["name"])
```

What prints? How does Python know which fact to grab?


```
pet = {"kind": "cat", "age": 2, "hungry": True}
print(pet["age"])
print(pet["hungry"])
```

What two lines print?


```
player = {"name": "Mia", "level": 3}  
print(player["score"])
```

There is no **"score"** key. What happens?

2 · Spot the Bug 🐛

Each snippet is broken. Read what it should do, then rewrite it so it works. Explain why the original was wrong.

Bug A — This should print the player's name. Instead it reads a number, and dictionaries are not read that way.

```
player = {"name": "Mia", "level": 3}
print(player[0])
```

Hint: dictionaries are read by **key name**, not by position number.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should make a dictionary, but it uses the wrong brackets, so it is really a broken list.

```
player = ["name": "Mia", "level": 3]
```

Hint: a dictionary is wrapped in curly braces `{ }`, not square brackets `[ ]`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should hold two facts about a pet, but a colon is missing, so it crashes.

```
pet = {"kind" "cat", "age": 2}
```

Hint: every key needs a  between it and its value.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this working dictionary carefully.

```
hero = {  
    "name": "Aria",  
    "hp": 100,  
    "weapon": "bow",  
}  
  
print(f"{hero['name']} has {hero['hp']} HP")  
print(f"Weapon: {hero['weapon']}")
```

What is **hero** ? How many key:value pairs does it hold?

What does **hero["name"]** read? What about **hero["weapon"]** ?

Why is **hero["hp"]** clearer than remembering that HP is "the second thing" in a list?

Submit 

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